

Quiz (Transactions and Locks III)

Instructions: For each problem, you must perform all of the following tasks correctly in order to get credit.

1. Circle all SQL commands that block.
2. Write an X through all commands that error.
3. If a command causes a deadlock, write DEADLOCK and stop the problem.
4. Write the output of each SELECT statement.

You will be allowed to use a computer and access the internet, but you will not be allowed to connect to postgres and run postgres commands.

Problem 1:

Session 1

```
1 CREATE TABLE t ( a INT );
2
3 VACUUM t;
4
5
6
7 VACUUM FULL u;
```

Session 2

```
1
2 CREATE TABLE u ( a INT );
3
4
5 VACUUM FULL u;
6 VACUUM t;
7
8 SELECT count(*) FROM t;
9 SELECT count(*) FROM u;
```

Problem 2:

Session 1

```
1 CREATE TABLE t (  
2     a INT PRIMARY KEY  
3     DEFERRABLE INITIALLY DEFERRED  
4 );  
5 BEGIN ISOLATION LEVEL  
6 REPEATABLE READ;  
7 INSERT INTO t VALUES (5);  
8  
9  
10 INSERT INTO t VALUES (6);  
11 SELECT count(*) FROM t;  
12  
13  
14 COMMIT;
```

Session 2

```
1  
2  
3  
4  
5  
6  
7  
8 BEGIN;  
9 INSERT INTO t VALUES (6);  
10  
11  
12 INSERT INTO t VALUES (5);  
13 ABORT;
```

Problem 3:

Session 1

```
1 CREATE TABLE t ( a INT );
2 INSERT INTO t VALUES (9);
3 INSERT INTO t VALUES (10);
4
5
6
7 UPDATE t SET a = a+1;
```

Session 2

```
1
2
3
4 BEGIN ISOLATION LEVEL
5 READ UNCOMMITTED;
6 SELECT count(*) FROM t;
7
8 DELETE FROM t WHERE a=10;
9 COMMIT;
10 SELECT count(*) FROM t;
```

Problem 4:

Session 1

```
1 CREATE TABLE t ( a INT );
2 INSERT INTO t VALUES (9);
3 INSERT INTO t VALUES (10);
4
5
6
7 UPDATE t SET a = a+1;
```

Session 2

```
1
2
3
4 BEGIN ISOLATION LEVEL
5 REPEATABLE READ;
6 SELECT count(*) FROM t;
7
8 DELETE FROM t WHERE a=10;
9 COMMIT;
10 SELECT count(*) FROM t;
```